# Siddhesh Bhadale

Backend & Platform Engineer | Distributed Systems, Networking & Real-Time Infrastructure

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## SUMMARY

Backend and platform engineer with 2 years shipping production systems for port, customs and enterprise customers. Technical lead for a multi-service remote-inspection platform — architecture, Node.js and Go services, MongoDB data model, and the Docker Swarm to Kubernetes migration — in a four-engineer team; it now runs at multiple port sites. Comfortable working a layer below the application: authored a Rust/libp2p peer-to-peer transport that lets those same services be reached from mobile and desktop clients with no exposed port, public IP or VPN. Looking for backend or platform engineering roles focused on distributed systems, networking, reliability and performance.

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Go, Rust

**Production exposure:** Kotlin, Swift

**Backend:** Node.js, NestJS, Express.js, REST API design, WebSockets, RPC over streams, Kong API Gateway, Nginx, microservices, modular monolith

**Clients:** React Native, React, Tauri

**Data:** MongoDB (schema design, indexing, aggregation pipelines), MySQL, Redis

**Infrastructure:** Docker, Docker Swarm, Kubernetes, Linux (Ubuntu/Debian), Nginx reverse proxy & TLS, Make and shell release automation, on-premises & air-gapped deployment

**Systems & Networking:** libp2p, Tokio, QUIC/TCP, NAT traversal (DCUtR hole punching, circuit relay), Noise, Yamux, UniFFI, WebRTC/WHIP, ONVIF/PTZ

**System Design:** Service decomposition, authentication & authorisation, RBAC, multi-tenancy, notifications, payments & subscriptions, logging, monitoring, performance profiling

**Practice:** Git, code review, Jira (epics, stories, sprint planning), technical leadership, R&D-driven design, stakeholder communication

## EXPERIENCE

**Junior Mobile Application Developer** Dec 2024 — Jun 2026

*EaseMyAI* *Thane, India*

- Technical lead for TeamStream, a remote-inspection platform for port and customs operations. Ran requirement discovery, system architecture, tech-stack selection and multiple R&D cycles, then built and shipped the backend, the React web app and the React Native mobile apps with a four-engineer team.
- Designed the backend as independently deployable Node.js and Go services behind Kong and Nginx — authentication, RBAC, session scheduling, email/push notification, recording and media-session orchestration — with the MongoDB schema, indexes and aggregation pipelines for session and event data.
- Owned deployment end to end: containerised every service, ran the platform on Docker Swarm, then migrated it to Kubernetes; automated image builds and rollouts with Make and shell scripts, turning a multi-step manual release into a single command.
- Supported UAT involving approximately **200 participants across 5 live-inspection sessions** with multiple live camera feeds each, on a single Mac mini M4 (24 GB). The platform is deployed at multiple port sites, and was showcased at the **India AI Impact Summit 2026**, where it was demonstrated to Ashwini Vaishnaw, Union Minister for Electronics & Information Technology.
- Built the Rust/libp2p connectivity layer (below) that made these on-premises deployments reachable from client apps without exposing ports or public IPs — unblocking rollout at sites where opening the network was not permitted.
- Led beyond the code: reviewed pull requests, planned epics, stories and sprints in Jira, unblocked teammates and ran daily coordination with project owners and stakeholders. Also delivered enterprise work for customers across aviation, ports and logistics — two React Native apps, an Electron desktop app and several web applications.

**Mobile Application Developer — Intern** Aug 2024 — Dec 2024

*Applied Cloud Computing* *Mumbai, India*

- Built production features for a top-10 Indian AMC's investor and distributor mutual-fund applications in React Native, integrating 34+ REST APIs for folio creation, SIP purchase and modification, and portfolio reporting.
- Integrated the Razorpay payment gateway across UPI, autopay, net banking and bank transfer flows, including failure and reconciliation paths.

**Full Stack Developer — Freelance** Dec 2023 — Jul 2024

*Independent* *Remote*

- Delivered web and mobile products for 20+ clients as sole engineer, covering design, backend, deployment and handover; the largest deployment serves 1,000 daily active users.

## SELECTED PROJECTS

**Peer-to-Peer Connectivity Layer for On-Premises Systems** *EaseMyAI | Sole engineer*

Rust, libp2p 0.56 (DCUtR, circuit relay, Noise, QUIC/TCP), Tokio, UniFFI, Node.js, Express, MongoDB, Docker, systemd, Kotlin, Swift

- Port, airport, defence and government customers run our video-management and inspection software strictly on-premises and will not expose a public IP, open a port or run a VPN. Built a libp2p transport in Rust that dials the on-prem server directly instead — about 19,600 lines across Rust, TypeScript, Kotlin and Swift, written solo.
- Reserved a circuit-relay path for guaranteed reachability, then upgraded to a direct connection via DCUtR hole punching where the NAT pair allowed it — degrading to relayed delivery on symmetric NAT and CGNAT rather than failing.
- Kept every existing service unchanged by exposing the tunnel as a loopback proxy on `127.0.0.1`, so REST, WebSocket, chat, file transfer and live video ran over P2P with no application rewrite. Hand-wrote the HTTP/1.1 proxy in Rust — hop-by-hop header handling, chunked encoding, CORS injection, reverse-proxy and CONNECT modes, TLS to self-signed LAN targets.
- Built the Node.js/Express/MongoDB bootstrap service behind it: server registration, heartbeat, connection-state lifecycle, short-lived session tokens and QR-code pairing for field onboarding. Instrumented it for operators with a live dashboard covering hole-punch success rate, relay circuits, ping RTT, per-peer state and traffic, plus webhook events for audit and automation.
- Shipped one Rust core to 4 Android ABIs and 3 iOS targets through UniFFI, with Docker, systemd and Make deployment tooling. Tested P2P connectivity with **40+ peer sessions** against a single on-prem server under a controlled test workload; the layer now backs two products — the inspection platform and an AI video-analytics VMS.

**Cross-Platform Product Suite (Mobile, Desktop, Web)** *Independent | Sole engineer*

NestJS, TypeScript, MongoDB, React Native, Tauri, Rust, React, Kotlin, Swift

- Designing and building a full product solo: a NestJS + MongoDB modular-monolith backend with clear module boundaries, serving a React Native mobile app, a Tauri/Rust desktop app and a React web client from one API surface.
- Wrote the platform-specific pieces natively — Kotlin and Swift on mobile, Rust on desktop — above a single shared TypeScript domain layer, and handled the cross-cutting concerns directly: RBAC, subscription and payment flows, notification delivery, and an offline-first sync model for clients that lose connectivity mid-session.

**Tamper-Proof Document Issuance & Verification** *Smart India Hackathon | Sole engineer*

Solidity, Ethers.js, MetaMask, Node.js, Express, MongoDB, React, React Native

- Built a blockchain-backed document system for the Bajaj Finserv Healthcare problem statement: hospitals issue a document, the holder receives a unique ID, and the insurer verifies it by ID plus emailed OTP — removing the file-upload step that made claim fraud possible. Wrote and deployed the Solidity contracts and the Node/Express and MongoDB services around them, and migrated the whole deployment from Goerli to Sepolia in about an hour the day before judging when Goerli was sunset without warning.

## OPEN SOURCE

- **jiangdongguo/AndroidUSBCamera #753** — *"Add 16 KB page size support for newer Android devices and rebuild native libraries."* Upstream pull request to a 2.7k-star, 930-fork UVC camera engine that has had no upstream commit since Sep 2024; opened from my fork on branch `feature/16kb-page-support`. The library gated the body-camera feature in our inspection platform.
- Upgraded Gradle 6.7.1 to 8.13 and moved to NDK 27, SDK 36 and Kotlin 2.0.21; dropped the prebuilt `.so` blobs and rebuilt `libuvc`, `libnative` and `libausbc` from source through `externalNativeBuild`, enabling `ANDROID_SUPPORT_16K_PAGES` for the 16 KB memory page size now required by Play Store, and used `jniLibs.pickFirst` to resolve duplicate-native conflicts.
- **+316 / −197 lines across 63 files in a single commit**; addresses 4 open upstream issues (#735, #744, #745, #746). Verified in a production Android app on Realme Narzo N55, Realme Narzo 80x, Samsung Galaxy S24 Ultra and Galaxy A15. Open and awaiting maintainer review.
- **Referenced downstream** by nolte/kamerplanter-android #19 (merged Aug 2026), which cites #753 as the upstream fix for the same 16 KB alignment failure and ships a maintained fork of the library because the upstream PR is still unmerged.

## EDUCATION

**B.E. Computer Engineering** 2020 — 2024

*Savitribai Phule Pune University (SPPU)* *Pune, India*

**Languages:** English, Hindi, Marathi